

SOUTH WOOD COUNTY AP, WI, USA

WMO: 726452

Lat: 44.359N

Lon: 89.837W

Elev: 1019

StdP: 14.16

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 04826

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.7	-6.1	-18.5	2.1	-9.0	-14.2	2.6	-5.0	21.5	14.4	19.6	15.6	4.9	290	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	89.6	72.7	86.2	70.6	83.4	69.2	75.4	84.7	73.6	82.3	71.9	80.2	10.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.7	125.7	80.0	70.8	117.9	78.3	69.4	112.1	77.0	39.6	85.1	37.8	82.3	36.2	79.8	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	18.1	16.3	-19.0	94.3	4.9	3.6	-22.5	96.9	-25.4	99.0	-28.2	101.0	-31.8	103.6
			-19.3	78.5	4.9	2.4	-22.9	80.2	-25.7	81.6	-28.4	82.9	-32.0	84.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	16.8	20.2	31.9	45.1	57.0	66.4	71.1	68.6	61.2	48.1	35.1	22.3	(6)
(7)		DBStd	21.21	12.35	11.89	11.92	9.57	8.62	6.60	5.83	5.60	8.15	9.28	9.89	11.49	(7)
(8)		HDD50	4146	1028	836	576	203	29	0	0	9	151	454	860		(8)
(9)		HDD65	7671	1493	1256	1028	600	276	61	14	27	166	530	896	1324	(9)
(10)		CDD50	2485	0	0	14	55	246	493	655	577	345	92	8	0	(10)
(11)		CDD65	537	0	0	1	2	28	104	204	139	53	6	0	0	(11)
(12)		CDH74	5164	0	0	8	46	357	1003	2010	1200	485	55	0	0	(12)
(13)		CDH80	1530	0	0	0	7	97	273	701	324	123	5	0	0	(13)
(14)	Wind	WSAvg	7.3	7.6	7.7	7.9	8.6	8.0	6.9	6.4	6.0	6.6	7.3	7.5	7.5	(14)
(15)	Precipitation	PrecAvg	31.6	0.9	0.9	1.8	3.0	3.9	4.6	3.8	3.9	3.4	2.5	1.7	1.3	(15)
(16)		PrecMax	41.2	2.5	2.1	5.3	5.5	8.5	9.4	9.8	10.0	8.5	5.9	4.6	3.2	(16)
(17)		PrecMin	24.4	0.3	0.3	0.2	1.1	1.4	1.7	0.8	0.6	0.6	0.3	0.0	0.2	(17)
(18)		PrecStd	3.8	0.5	0.4	1.1	1.2	1.4	2.2	2.1	2.0	1.9	1.3	1.0	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.0	53.9	72.5	80.5	88.2	90.5	95.5	91.0	88.4	80.3	66.8	54.2	(19)
(20)		MCWB	38.9	48.1	60.0	61.1	68.8	73.3	75.9	73.5	72.2	66.2	55.3	50.5		(20)
(21)		2%	DB	40.2	45.2	62.9	72.6	82.1	86.2	90.4	87.2	83.6	74.1	59.7	45.4	(21)
(22)		MCWB	36.0	39.6	52.1	55.9	64.8	70.2	73.6	71.6	69.7	62.8	52.3	40.7		(22)
(23)		5%	DB	36.1	39.5	55.3	67.5	77.8	82.6	86.9	83.3	79.6	69.3	54.8	40.5	(23)
(24)		MCWB	33.5	35.6	47.7	52.9	62.3	68.5	71.4	69.9	66.4	60.0	48.0	37.1		(24)
(25)		10%	DB	33.7	36.1	49.0	62.7	72.9	80.2	83.7	80.8	75.4	63.8	50.1	36.6	(25)
(26)		MCWB	31.9	33.0	41.8	49.5	59.4	66.8	69.5	67.9	64.6	56.5	44.6	33.9		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.6	46.4	60.7	63.4	71.7	76.3	78.9	77.1	74.2	69.8	57.8	51.3	(27)
(28)		MCDWB	44.6	52.6	71.0	74.3	83.7	86.1	90.2	85.7	84.4	77.0	63.9	53.2		(28)
(29)		2%	WB	36.3	39.9	54.9	58.6	67.9	73.8	76.1	74.5	71.6	65.1	53.5	42.3	(29)
(30)		MCDWB	39.7	44.5	60.0	69.4	77.4	83.0	85.7	82.7	80.1	70.5	58.7	44.4		(30)
(31)		5%	WB	33.8	36.1	48.1	54.6	64.8	71.5	74.2	72.3	69.4	61.2	48.7	37.3	(31)
(32)		MCDWB	35.7	40.0	56.1	64.5	73.8	79.7	83.1	79.9	75.6	67.8	53.9	40.4		(32)
(33)		10%	WB	31.5	33.0	42.2	50.5	62.1	68.9	72.2	70.3	67.0	57.1	44.9	34.1	(33)
(34)		MCDWB	33.6	35.7	48.6	60.6	69.9	76.2	80.6	77.6	73.7	63.3	49.3	36.3		(34)
(35)	Mean Daily Temperature Range		MDBR	15.5	17.0	18.6	21.3	22.1	21.2	21.9	20.8	21.7	19.1	15.9	14.1	(35)
(36)		5% DB	MCDBR	16.8	18.8	25.8	30.5	27.7	24.9	24.6	23.5	24.9	25.5	24.0	17.8	(36)
(37)		MCWBR	13.7	14.0	16.1	16.6	13.1	11.7	10.2	10.3	12.2	15.0	16.5	14.3		(37)
(38)		5% WB	MCDBR	15.7	17.9	23.9	26.2	23.6	21.7	21.3	19.8	20.6	22.1	22.0	15.5	(38)
(39)		MCWBR	13.4	13.4	15.9	16.0	12.6	11.7	10.6	10.0	12.0	15.0	16.8	13.2		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.254	0.285	0.320	0.354	0.389	0.394	0.385	0.388	0.358	0.321	0.289	0.262	(40)	
(41)		taud	2.402	2.311	2.376	2.360	2.310	2.325	2.356	2.348	2.428	2.521	2.527	2.452	(41)	
(42)		Ebn at Noon	281	288	291	288	280	278	279	274	275	273	266	267	(42)	
(43)		Edh at Noon	23	31	33	37	40	40	38	37	31	25	21	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	543	896	1247	1436	1716	1885	1936	1653	1314	834	533	397	(44)	
(45)		RadStd	62	80	140	142	131	141	97	109	97	95	54	45	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	-3.96	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page